

AI Auto-Invoicing Product – Project Understanding

The client wants to build an **AI-powered Auto-Invoicing System** that can automate the manual invoicing process of their business.

Main Objective

The goal is to automatically collect business/transaction data, process it, generate accurate invoices, validate them, and potentially send them to customers without requiring manual work at every step.

Possible Workflow

Business Data / Orders / Transactions

↓

AI Processing & Data Extraction

↓

Invoice Information Generation

↓

Validation & Calculation

↓

Automatic Invoice Generation

↓

Database Storage

↓

Send Invoice to Customer

Key Features

1. Data Collection

- Customer details
- Products or services
- Quantity
- Price
- Tax/GST
- Payment information
- Due date

2. AI-Based Data Processing

- Extract information from emails, PDFs, documents, receipts, or other unstructured data if required.
- Identify customer details, products, prices, taxes, and total amounts.

3. Automatic Invoice Generation

- Generate invoice numbers
- Add customer information

- Add product/service details
- Calculate taxes
- Calculate total amount
- Generate PDF invoices

4. Invoice Validation

- Verify calculations
- Validate tax amounts
- Check for missing information
- Detect duplicate invoices
- Ensure invoice data is accurate

5. Automation

- Automatically generate invoices
- Store invoices in a database
- Send invoices to customers through email or APIs
- Reduce manual invoicing work

6. AI Integration

- Use LLMs for understanding unstructured business data.
- Use OCR/document processing when information comes from scanned documents or images.
- Combine AI with traditional business rules for better accuracy and reliability.

Possible Backend Architecture

FastAPI Backend

- Invoice Processing Service
- OCR / LLM Data Extraction
- Validation & Calculation Engine
- Invoice Generator
- MySQL/PostgreSQL Database
- Email/API Integration

Important Point

The job description is **high-level**, so the exact requirements are not clearly mentioned. Before development, the client should clarify:

- Where will the invoice data come from?
- Emails, PDFs, Excel files, orders, database, or manual input?
- Which accounting/payment system needs to be integrated?

- Is GST/VAT/tax calculation required?
- Should invoices be generated as PDFs?
- Should invoices automatically be emailed to customers?
- Does the system need OCR?
- Which LLM/AI capabilities are actually required?